

Hidden keys and repeated biometric records

Can multiple protected records reveal identity when keys remain hidden?

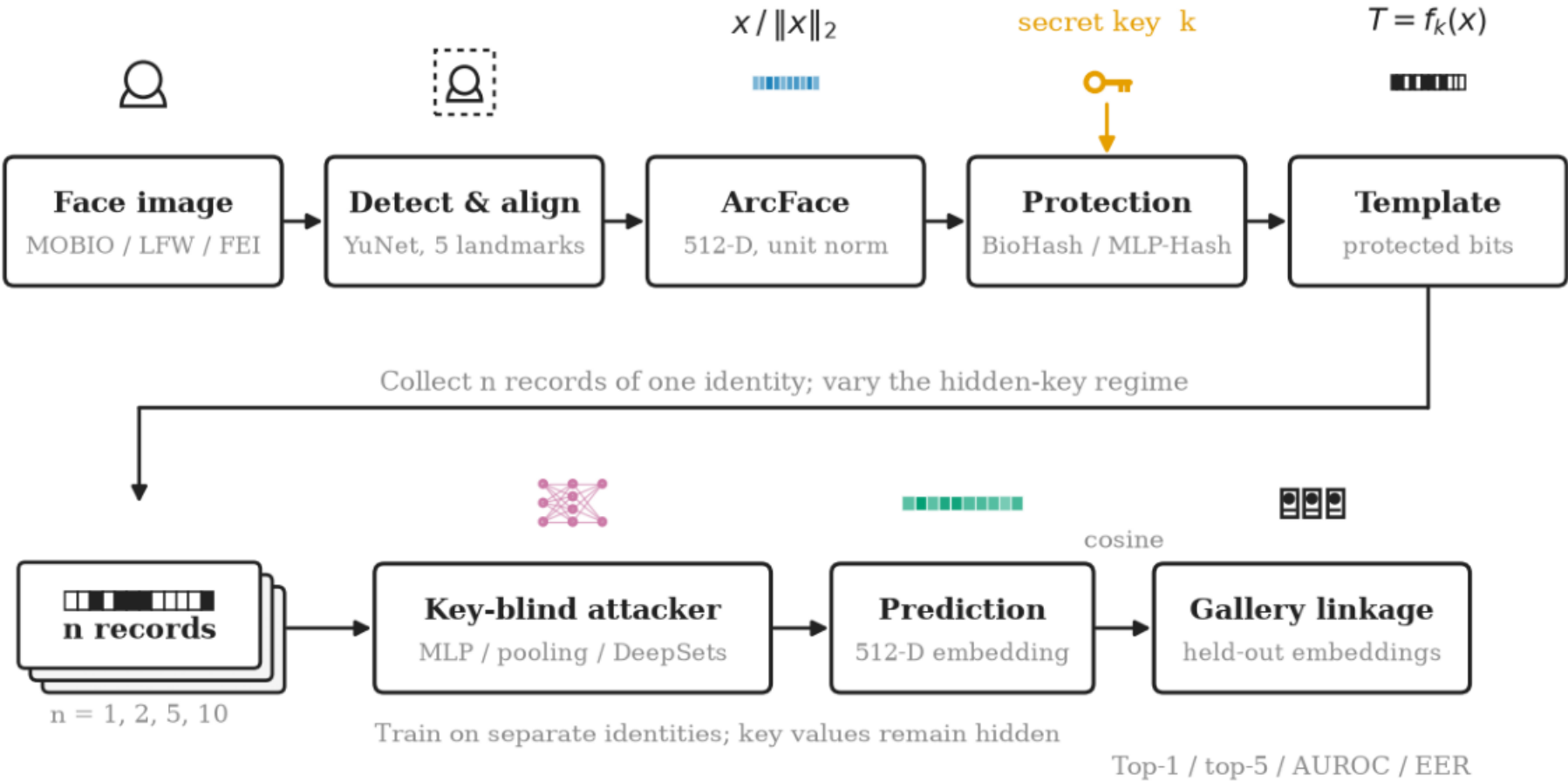
Fresh independent keys: chance-compatible results in the tested attacks.

Recurring transforms: large gains from multiple same-identity records.

Boundary location changes with the dataset and identity partition.

Independent study. No exact benchmark reproduction or universal privacy claim.

Experimental architecture



Train on separate identities. Reconstruct an embedding, then link to a held-out gallery.

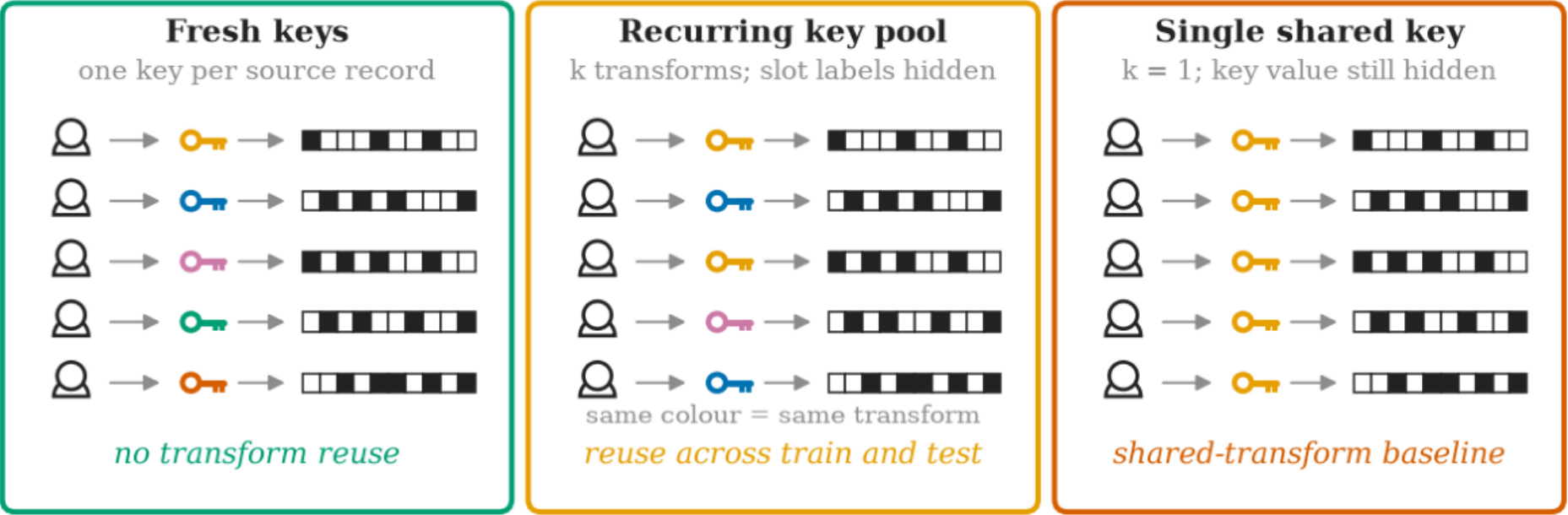
Dataset and protection coverage

Dataset	Identities	Train/val/test	Embeddings	Variation
MOBIO	150	90/30/30	1,799 / 1,800	Sessions
LFW	125	75/25/25	1,500 / 1,500	In-the-wild
FEI	200	120/40/40	2,378 / 2,400	Pose/expression

BioHash: 128 bits; all three datasets.
MLP-Hash: 512 bits; MOBIO only; paper-specified, not source-exact.
Haar-sign-corrected BioHash: MOBIO implementation control.

FEI: 22 low-illumination failures; all identities retain a gallery image plus 10 exposures.

What changes between key regimes?

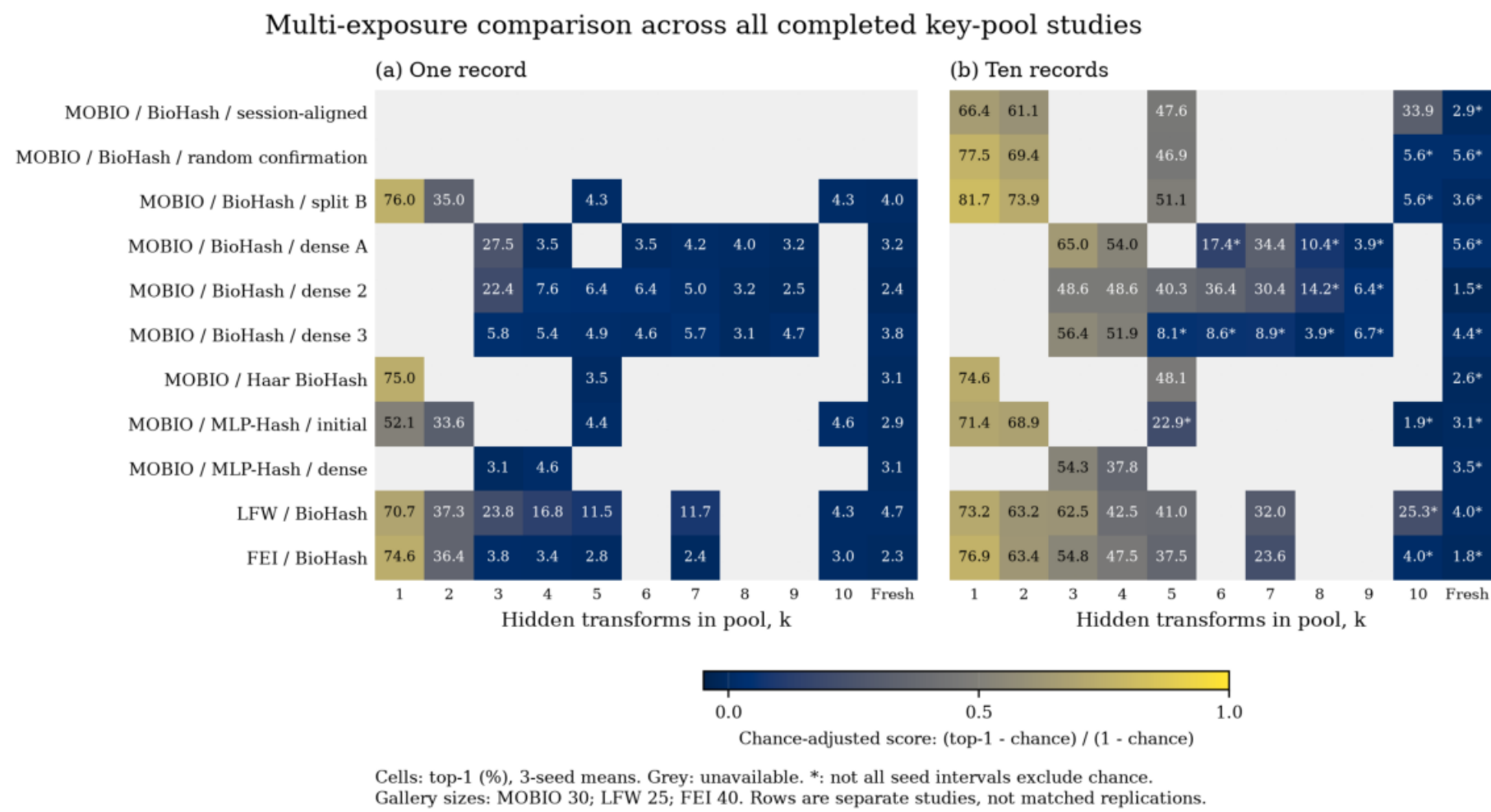


Chance-compatible results **Multi-record gains observed** **Strong linkage observed**
ideal invariance needs assumptions boundary depends on dataset and split hidden does not mean unlearnable

Protocol Disjoint train / validation / test identities; one held-out gallery image per identity
8 nested sets per identity; 3 model seeds; identity-clustered bootstrap intervals

Key values stay hidden in every regime. Recurring transforms are shared across identity splits.

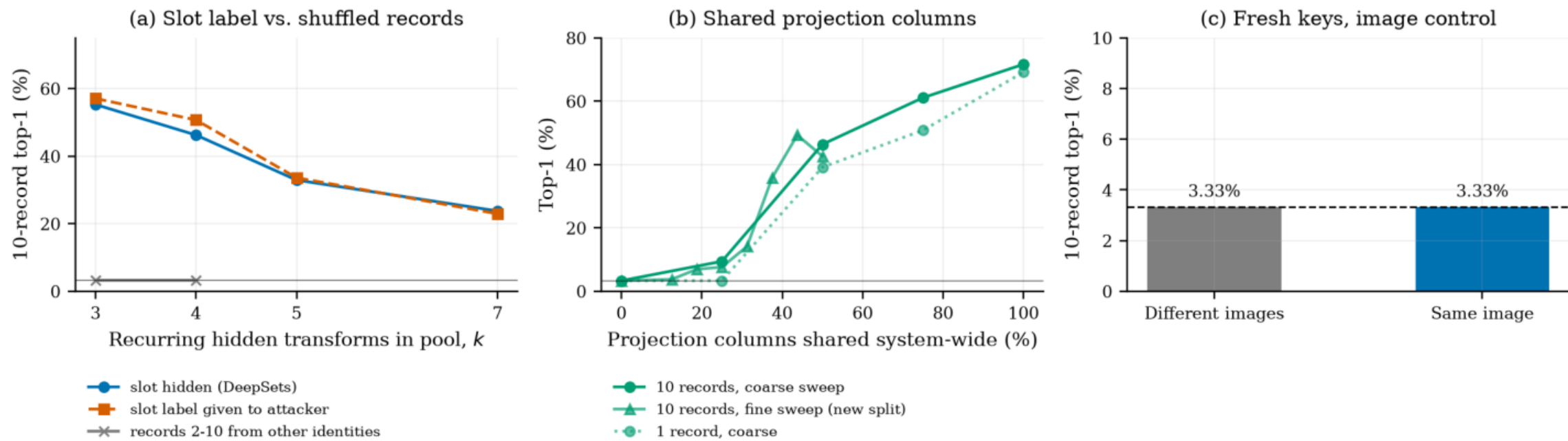
All completed key-pool studies



65 conditions from 11 studies. Missing endpoints stay blank; interval failures remain visible.

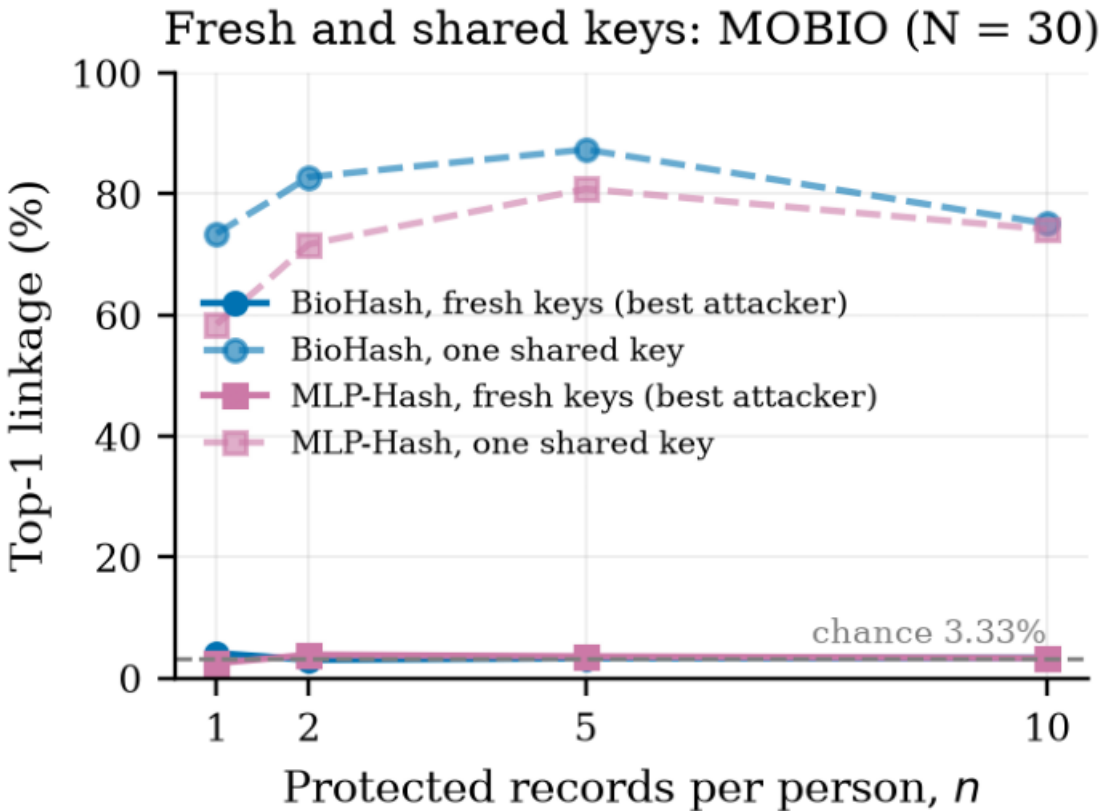
Mechanism and correlation controls

MOBIO mechanism controls (30 gallery identities; chance 3.33%)



Slot identifiers are not key values. Coarse and fine correlation sweeps use separate partitions.

Fresh-key endpoints across datasets



BioHash, 10-record mean-pool endpoint:

MOBIO: 3.61% (chance 3.33%)

LFW: 4.00% (chance 4.00%)

FEI: 1.77% (chance 2.50%)

**Chance inclusion is not equivalence.
No universal privacy guarantee.**

Ready to draft; not yet ready to submit

Completed: three datasets, two schemes on MOBIO, reuse and mechanism controls.

Before submission:

1. Approve and obtain the next dataset; approve two additional schemes.
2. Freeze the extension protocol, then run pilots and confirmation.
3. Define equivalence margins; strengthen paired uncertainty analysis.
4. Review theory, finite-key assumptions, norm controls and novelty.
5. Obtain Sani's scientific and presentation review.

A/A* is a venue ambition, not an established property or an acceptance prediction.